

One-Component Stereoscopic Oligosiloxane–Based Polyurea Radiative Cooling Coatings

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Passive daytime radiative cooling (PDRC) technology is an emerging and promising zero-carbon solution. However, existing PDRC materials usually exhibit poor mechanical properties and lack self-cleaning ability, which can reduce the lifespan of the coating and affect its cooling effect. To address these challenges, a one-component stereoscopic oligosiloxane–based polyurea radiative cooling coating is developed. The coating is fabricated by integrating blocked amine–oligosiloxane nanoclusters, isocyanate-terminated curing agents, and inorganic aluminum oxide (Al_2O_3) particles. The unique stereoscopic oligosiloxane structure, characterized by high functionality and self-reinforcing effect, endows the polyurea with a highly cross-linked organic–inorganic hybrid network structure. This structure provides exceptional hardness and modulus for the coating, with a surface hardness of ≈ 227 Mpa and a modulus of ≈ 9.7 Gpa. In addition, the high infrared emissivity of Si–O–Si bonds and the high reflectivity of Al_2O_3 enable the coating to have an excellent cooling effect of $\approx 9^\circ\text{C}$ below ambient temperature during the daytime. The coating also exhibits excellent versatility. The introduction of polydimethylsiloxane (PDMS) endows the coating with excellent self-cleaning and antifouling properties. Furthermore, spraying fluorinated nanosilica (F-SiO_2) on the surface enables superhydrophobicity and water droplet power generation. The novel PDRC coating provides a sustainable solution for cooling buildings.

1. Introduction

Driven by the global sustainable economy and carbon neutrality, energy consumption in buildings, particularly cooling demand, has become a critical concern. Currently, air conditioning is widely used as a traditional cooling method to provide comfort during the hot summer months. However, greenhouse gas emissions from traditional cooling systems intensify the greenhouse effect and contribute to climate change, posing a significant threat to the environment.^[1] In recent years, passive daytime radiative cooling (PDRC) has gained significant attention

as a low-energy, eco-friendly cooling solution.^[2] It works by radiating heat from Earth into the cold outer space through the atmospheric transparency window, which allows the passage of electromagnetic waves in the 8–13 μm range.^[3] The enormous temperature difference between Earth (≈ 300 K) and the cold universe (≈ 3 K) allows heat to flow spontaneously from hot to cold spaces, leading to a cooling effect.^[4] This technology provides a sustainable and effective solution for cooling building spaces.

PDRC materials are primarily classified into three categories: inorganic, organic, and organic–inorganic. Inorganic materials such as silica (SiO_2),^[5] alumina oxide (Al_2O_3),^[6] and barium sulfate^[7] are known for their strong heat emission and low solar absorptivity. These properties are attributed to phonon-polariton resonance, wide bandgap, and high refractive index.^[8] However, inorganic materials generally require high-temperature sintering and complex fabrication processes, and they usually exhibit poor toughness, making them not well-suited for coating applications.

Organic materials are highly recommended for use in PDRC because of their low extinction coefficients (k) in the solar spectrum.^[9] Polymers such as polydimethylsiloxane (PDMS),^[10] poly(vinylidene fluoride-co-hexafluoropropene),^[11] polyvinylidene fluoride,^[12] polyethylene terephthalate,^[13] and polymethyl methacrylate^[14] are excellent candidates for radiative cooling because they minimize solar heat gain. They exhibit a desirable selective absorption band overlapping with the atmospheric transparency window because of the molecular vibrations of the Si–O–Si, C–C, C–F, and C–H bonds.^[15] However, organic materials generally exhibit low solar reflectivity, which affects their cooling ability. To address these challenges, organic–inorganic composites have emerged as a promising direction for radiative cooling in buildings. These composites combine the advantages of organic and inorganic materials and usually mix inorganic particles with high-emissivity polymers.^[8] However, most polymers exhibit poor mechanical properties (e.g., low hardness and modulus), making composites prone to fracture, detachability, and monofunctionality.^[16] These shortcomings hinder their ability to withstand harsh outdoor conditions. Designing a high-performance polymer that can be combined with inorganic

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particles to achieve excellent cooling and mechanical properties remains a significant challenge.

Herein, we designed a one-component stereoscopic oligosiloxane-based polyurea PDRC coating. The coating comprises a highly cross-linked organic–inorganic hybrid polyurea network and Al_2O_3 particles. First, hyperbranched amine–oligosiloxane nanoclusters (HASi) with high functionality and self-reinforcing properties were synthesized using hydrolysis-condensation reactions. The HASi were then blocked with isobutyraldehyde (B-HASi) and mixed with an isocyanate-terminated curing agent to form a one-component stereoscopic oligosiloxane-based polyurea. This structure provides a coating with high hardness and modulus. In addition, the introduction of Al_2O_3 particles enhances the coating emissivity and reflectivity. The combination of these materials results in a coating with an elastic modulus of ≈ 9.7 GPa and a surface hardness of ≈ 228 MPa. The coating achieved a solar reflectivity of ≈ 0.95 and an average atmospheric window emissivity of ≈ 0.90 . Under direct sunlight (656 W m^{-2}), the coating exhibited excellent PDRC performance, with an average temperature drop of $\approx 9^\circ\text{C}$ and a maximum temperature drop of $\approx 15^\circ\text{C}$. The coating also exhibits excellent versatility. The addition of PDMS as a lubricant imparts the self-cleaning and antifouling properties of the coating, preventing the accumulation of dust and other particulate contaminants on the surface. This helps maintain the radiative performance of the coating and enhances its outdoor durability. In addition, fluorinated nanosilica (F-SiO_2) was sprayed on the surface to impart the superhydrophobicity of the coating. This feature not only repels water but also enables the coating to generate electrical energy from rainwater by harnessing the potential energy of water droplets. The PDRC coating exhibits superior mechanical properties and environmental durability. This study provides a new perspective on the development of PDRC materials for cooling buildings.

2. Results and Discussion

2.1. Preparation of One-Component Stereoscopic Oligosiloxane-Based Polyurea Radiative Cooling Coatings

A highly cross-linked organic–inorganic hybrid network polyurea was prepared to enhance the mechanical properties of the polymer. First, hyperbranched HASi were synthesized using hydrolysis-condensation reactions (Figure 1a). The reaction between $-\text{NCO}$ and $-\text{NH}_2$ groups is extremely fast, leading to gelation within seconds.^[17] To prevent premature gelation, isobutyraldehyde was used as a sealant to react with HASi to block the amino groups (B-HASi). When HASi was added dropwise to the EA solution of HDI, it was rapidly gelatinized within 3 s (Figure S1, Supporting Information). When B-HASi was added to the EA solution of HDI, the mixture was not cure without exposure to moisture. Then, different mass fractions of Al_2O_3 particles were mixed with B-HASi and the hexamethylene diisocyanate to fabricate the PDRC coatings. The mixture was then applied to a concrete plate and cured at room-temperature (Figure 1b). The Si–O–Si bonds in the network exhibit strong absorption properties in the fingerprint region ($1500\text{--}1600 \text{ cm}^{-1}$).^[18] According to Kirchhoff's law, strong absorption implies strong emission.^[19] Al_2O_3 particles were selected because of their wide electronic

bandgap (8.7 eV) and high refractive index ($n \approx 1.77$),^[20] which provide excellent solar reflectivity (Figure 1c). The coatings were designated as $\text{PU}_a\text{-x}$, where x represents the mass fraction of Al_2O_3 .

Nuclear magnetic resonance spectroscopy was used to confirm the successful synthesis of HASi (Figure S2, Supporting Information). Before HASi was blocked, the N–H stretching vibration caused a band at 3350 cm^{-1} (Figure S3, Supporting Information). This band disappeared after HASi blocking, and the C=N stretching vibration band emerged at 1670 cm^{-1} . The asymmetric stretching vibrations of the linear and branched Si–O–Si bonds peaked at 955 cm^{-1} . The Fourier transform infrared spectrum (FTIR) of $\text{PU}_a\text{-0}$ indicated that the blocked amino group was deblocked by moisture (Figure S4, Supporting Information).^[21] Phenyl groups were incorporated to enhance the toughness of $\text{PU}_a\text{-0}$ via spatial site resistance effects, with a phenyl skeleton stretching vibration peak at 1566 cm^{-1} . The quantitative FTIR spectra show that the proportion of H-bonded C=O in $\text{PU}_a\text{-0}$ was 72% (Figure S5, Supporting Information). The average DMT modulus of $\text{PU}_a\text{-0}$ was ≈ 2.7 GPa because of the rigid Si–O–Si core of the stereoscopic oligosiloxane and the high cross-linking density in the polyurea network (Figure S6, Supporting Information). Compared with previously reported polyurea-based radiative cooling coatings, where the polyurea networks with low cross-linking density inherently limit their mechanical properties,^[22] the one-component stereoscopic oligosiloxane-based polyurea coating demonstrates significant advantages. The unique stereo-oligosiloxane structure endowed the polyurea with a highly cross-linked organic–inorganic hybrid network structure. Thus, the coating exhibited superior mechanical properties and was able to withstand harsh outdoor conditions. In addition, the coating exhibited excellent versatility. The introduction of PDMS imparted the self-cleaning properties of the coatings. This method is straightforward and has the potential to produce PDRC materials on a large scale (Figure 1d).

2.2. Morphology and Properties of the Coatings

The scanning electron microscope (SEM) images of the $\text{PU}_a\text{-x}$ coatings are shown in Figure 2a. The surface roughness of the coatings increased with increasing Al_2O_3 content. The load-displacement curves obtained from the nanoindentation tests are shown in Figure 2b. The diamond tip was indented into different hybrid coatings at a fixed loading force (10 mN). Under the same load, coatings with higher hardness contents exhibited shallower indentations. The $\text{PU}_a\text{-x}$ coatings exhibited exceptional mechanical properties, including significant hardness and elastic modulus. As shown in Figure 2c, the hardness of coatings initially increased and then decreased with increasing Al_2O_3 content. The strength of coatings also exhibited a similar trend. The surface hardness and elastic modulus reached a maximum value when the mass fraction of Al_2O_3 is 70%. The $\text{PU}_a\text{-70}$ achieved optimal mechanical performance with a surface hardness of 228 ± 20 MPa and an elastic modulus of 9.7 ± 0.3 GPa. The superior mechanical properties of coatings can be attributed to the hyperbranched structure of B-HASi, which provides more interaction and reaction sites with other components. The addition of Al_2O_3 particles further enhanced the mechanical performance.

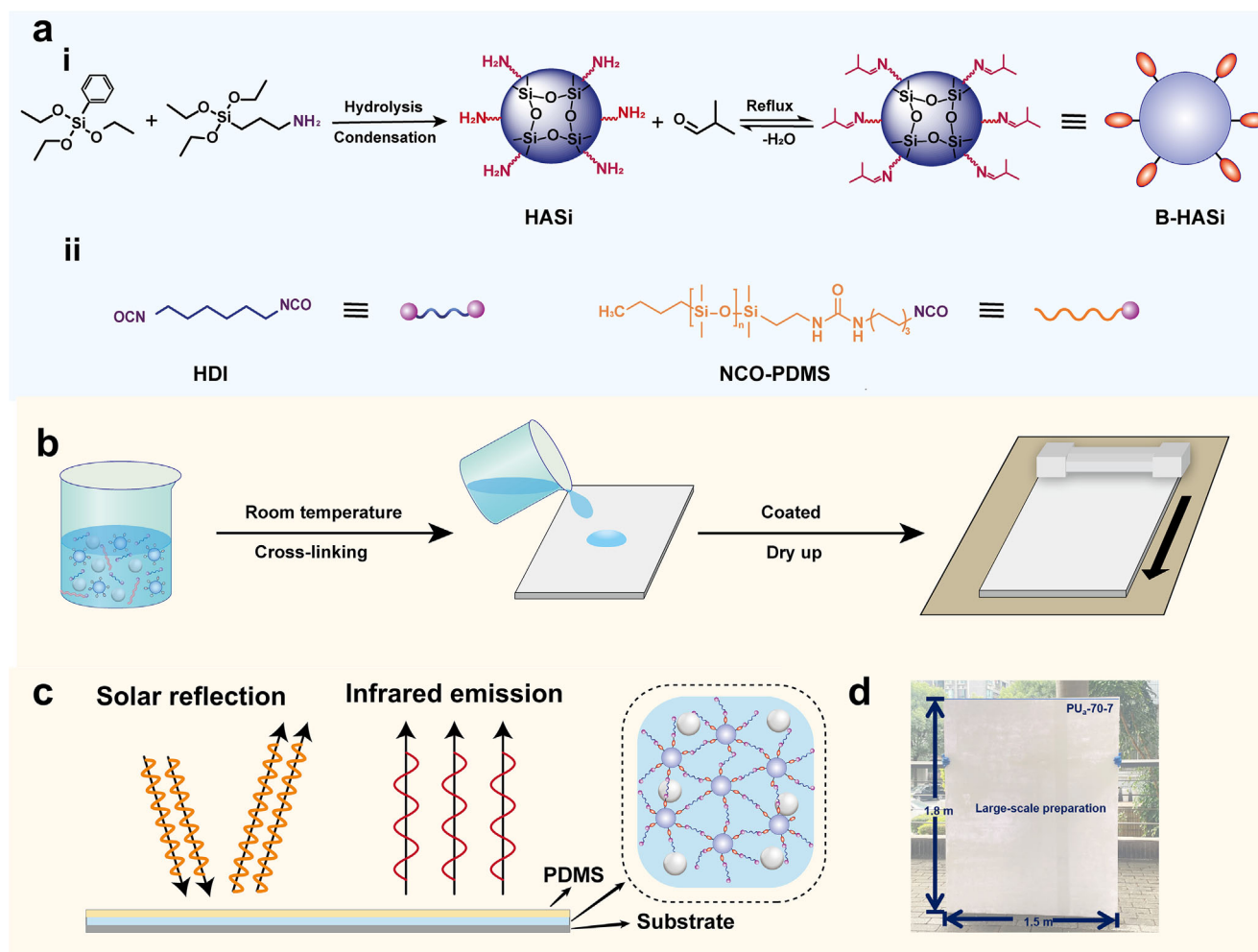


Figure 1. a) Synthesis of blocked HASi and the structure of the isocyanate-terminated curing agents. b) Preparation of one-component stereoscopic-based oligosiloxane polyurea radiative cooling coatings. c) Cooling mechanism of the coatings. d) Large-scale production display.

However, excessive Al₂O₃ content negatively affects the mechanical strength of the coatings, as seen in PU_a-80, which exhibits a reduced elastic modulus of 8.3 ± 0.6 GPa and a surface hardness of 179.0 ± 11 MPa. Shore hardness tests also showed that PU_a-70 exhibited the best mechanical properties (Figure S7, Supporting Information). The coating exhibits excellent abrasion resistance compared to commercial coatings (Figure 2d). The epoxy (EP) coatings exhibited higher abrasion values than PU_a-0 after 1000 wear cycles with a total load of 1 kg. Although the abrasion value increased with the addition of Al₂O₃ (PU_a-Al₂O₃), it remained significantly lower than that of the commercial polyacrylic acid (PAA) and polyurethane (PU) coatings. Thermogravimetric analysis was used to confirm the excellent thermal stability of the PU_a-x coatings (Figure S8, Supporting Information).

The solar reflectance (R_{solar} , 0.3–2.5 μm) and long-wave infrared (LWIR) emittance (ϵ_{LWIR} , 8–13 μm) of the coatings are shown in (Figure 2e,f). The results demonstrate that the addition of Al₂O₃ significantly enhances both R_{solar} and ϵ_{LWIR} . The particle size distribution analysis revealed that Al₂O₃ particles were primarily concentrated at ≈ 1.6 μm , with a smaller fraction ranging from 2 to 8 μm (Figure S9, Supporting Information). This broad

particle size distribution facilitates effective Mie scattering at the air-polymer interfaces, contributing to high solar reflectivity.^[23] The robust emissivity is attributed to the phonon-polariton resonance characteristics of Al₂O₃ particles.^[24] When the Al₂O₃ content exceeded 70%, the coatings achieved an average R_{solar} of 0.91 and ϵ_{LWIR} of 0.88 (Figure 2g). The optimal mechanical and optical properties of the coating were achieved at an Al₂O₃ mass fraction of 70%.

2.3. Introduction of NCO-PDMS

Self-cleaning property refers to the ability of a material to automatically remove contaminants under external forces, which is crucial for extending the outdoor service life of coatings.^[25] To impart coatings with remarkable hydrophilic and self-cleaning properties, isocyanate-terminated PDMS (NCO-PDMS) was introduced. This modification was applied to the PU_a-70 coating. The successful synthesis of NCO-PDMS was confirmed by FTIR spectra (Figure S10, Supporting Information). To enhance the self-enrichment of PDMS on the coating surface, NCO-PDMS

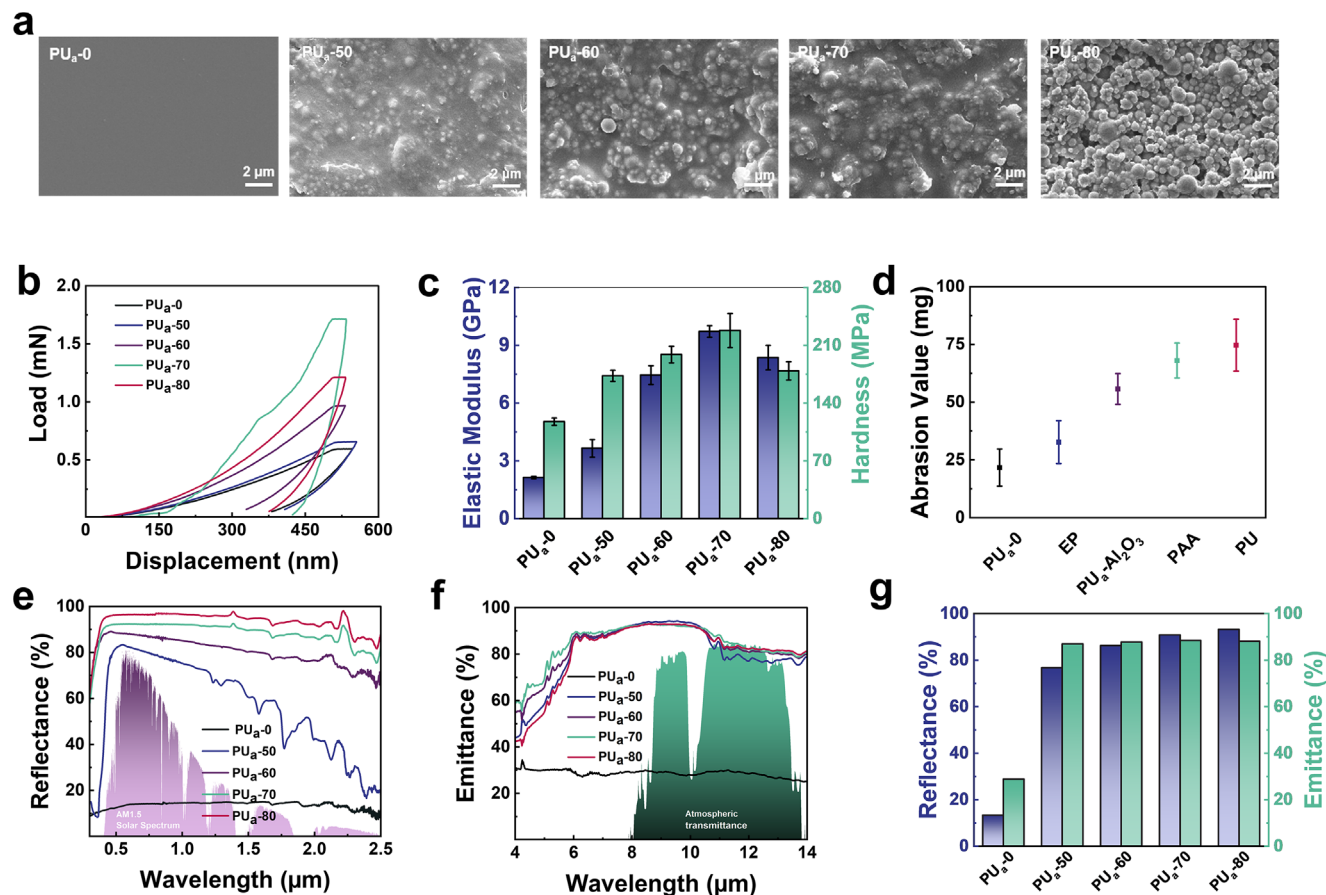


Figure 2. a) SEM image of the surface, b) load-displacement curves, and c) hardness and elastic modulus of the PU_a-x coatings. d) Comparison of the abrasion values between the PU_a-x coatings and commercial epoxy/ polyacrylic acid /polyurethane coatings. e) Reflectance spectrum in the solar radiation range along with the normalized ASTM G173 global solar spectrum. f) Emissance spectrum in the LWIR range and the atmospheric transmission window spectrum. g) Average reflectivity and emissivity of the PU_a-x coatings (reflection wavelength range: 0.3–2.5 μm; atmospheric LWIR transparent window: 8–13 μm).

was introduced at mass fractions of 1%, 3%, 5%, 7%, and 9%. The resulting coatings were designated as PU_a-70-y, where y represents the mass fraction of the NCO-PDMS.

The changes in the water contact angle (WCA) of the coatings are shown in Figure 3a,b. Compared with PU_a-70 (94° ± 5°), the introduction of NCO-PDMS significantly increased the WCA. The WCA values for PU_a-70-1, PU_a-70-3, PU_a-70-5, PU_a-70-7, and PU_a-70-9 were 103° ± 3°, 115° ± 4°, 122° ± 1°, 128° ± 4°, and 132° ± 2°, respectively (Figure 3b). Shore hardness tests revealed a gradual decline in surface hardness with increasing NCO-PDMS content (Figure S11, Supporting Information). When NCO-PDMS is added at more than 7%, the Shore hardness of the coating decreases to below 85 HD. In the trade-off between the Shore hardness and WCA, a 7% addition of NCO-PDMS was selected. The SEM image of PU_a-70-7 in Figure 3c reveals that PDMS is significantly enriched on the coating surface, which also affects the optical properties. The PU_a-70-7 coating achieved an average R_{solar} of 0.95 and ϵ_{LWIR} of 0.90 (Figure 3d–f), demonstrating significant improvements compared with PU_a-70 (R_{solar} of 0.91, ϵ_{LWIR} of 0.88).

To assess the solar reflectance and thermal management performance of the PU_a-70-7 coating, a xenon lamp with a uniform

power density of 1000 W m⁻² was used to simulate solar irradiance. An infrared camera was used to capture the infrared images and temperature data (Figure 3g–i). During the 12 min test, the temperature of the PU_a-70-7 increased from 25 to 35.2 °C. The result is significantly lower than that of the bare concrete plate (49.4 °C) and one coated with commercial white paint (38.1 °C). This outstanding performance is primarily attributed to the heat-radiating ability of the stereoscopic oligosiloxane-based polyurea and the excellent reflecting ability of Al₂O₃ particles.

2.4. Cooling Performance Assessment

Outdoor radiative cooling performance is a crucial factor in practical applications. The calculation of the $P_{\text{cooling, net}}$ using a radiative cooling model under various environmental conditions enables a quantitative analysis of the cooling performance of the PU_a-70-7. Assuming that an ambient temperature (T_{ambient}) of 25 °C (298 K), solar irradiance (I_{solar}) of 1000 W m⁻², and nonradiative heat exchange coefficient (hc) of 0, 3, 6, 9 and 12 W m⁻² K⁻¹. Under typical daytime conditions, the maximum $P_{\text{cooling, net}}$ of the coating reaches 50.8 W m⁻² (Figure 4a). To

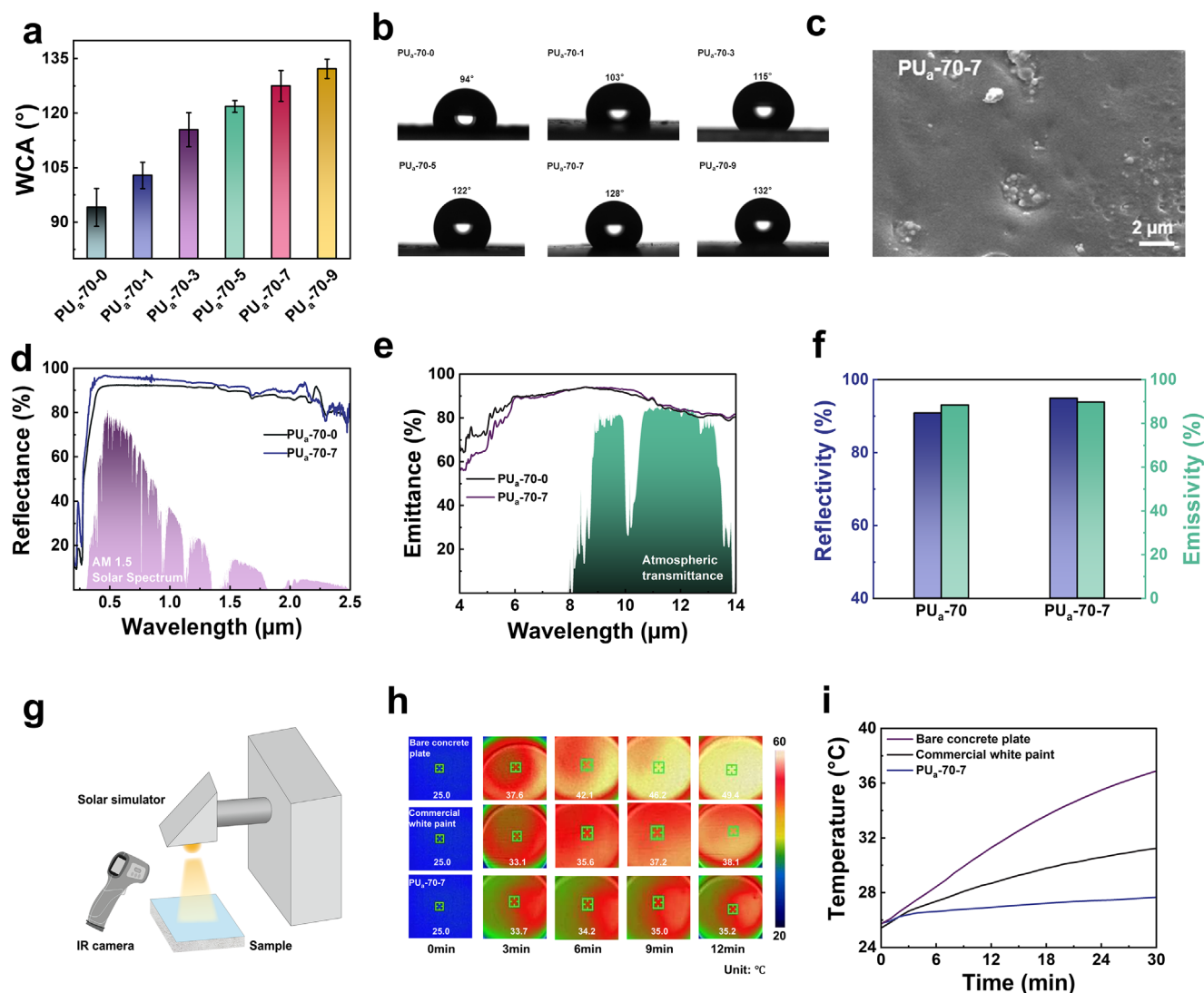


Figure 3. a) Water contact angle and b) water contact angle images of the PU_a-70-*y* coatings. c) SEM image of the surface of the PU_a-70-*y* coatings. d) Reflectance, e) emittance, and f) average reflectivity and emissivity of the PU_a-70-0 and PU_a-70-7 coatings. g) Schematic of the simulated solar heating test. h) Thermal infrared imaging and i) temperature monitoring of the bare concrete plate, commercial white paint, and PU_a-70-7, respectively.

comprehensively evaluate the PDRC performance of the coatings, tests were conducted in downtown Guangzhou (Longitude: 113.34°, Latitude: 23.15°) under two typical weather conditions: a cloudy day (October 21, 2024) and a sunny day (October 28, 2024). The tests were conducted at midday (11:00–13:00) when the solar irradiance reached its peak. The experimental setup for the outdoor cooling performance tests is illustrated in Figure 4b,c, and Figure S12 (Supporting Information). The test box was placed in an open area of the building to minimize interference from the surrounding structures and trees. The entire setup was elevated to a height of 1 m above the ground to prevent heat transfer between the ground and the equipment during testing. Under sunny conditions, with a solar irradiance (I_{solar}) of 656 W m⁻² and an average relative humidity (RH) of 27%, the coating achieved an average sub-ambient temperature drop ($\Delta T_{\text{cooling}}$) of ≈ 9 °C, reaching a maximum temperature difference of ≈ 15 °C (Figure 4d–f). Although commercial white coatings exhibit cooling effects, they

are less effective than the PU_a-70-7 coating in terms of reducing temperatures. The consistently higher temperatures of the bare concrete plate can be attributed to its lower solar reflectivity, which causes it to absorb excess heat rather than dissipate it through thermal radiation. Under cloudy conditions with a solar irradiance (I_{solar}) of 195 W m⁻² and an average RH of 43.8%, the coating achieved an average sub-ambient temperature drop ($\Delta T_{\text{cooling}}$) of ≈ 7 °C, with a maximum temperature reduction of ≈ 13 °C (Figure 4g–i). The cooling performance varies between sunny and cloudy conditions, primarily due to differences in atmospheric humidity and solar irradiance.^[26] Cloudy conditions exhibit higher humidity and water vapor content than sunny conditions, which affects the atmospheric infrared absorbance and emissivity in the 8–13 μm long wavelength range.^[27] In sunny conditions, direct sunlight provides heat gain, and variations in the intensity of direct sunlight correspond to variations in temperature reduction. These outdoor experiments demonstrate that

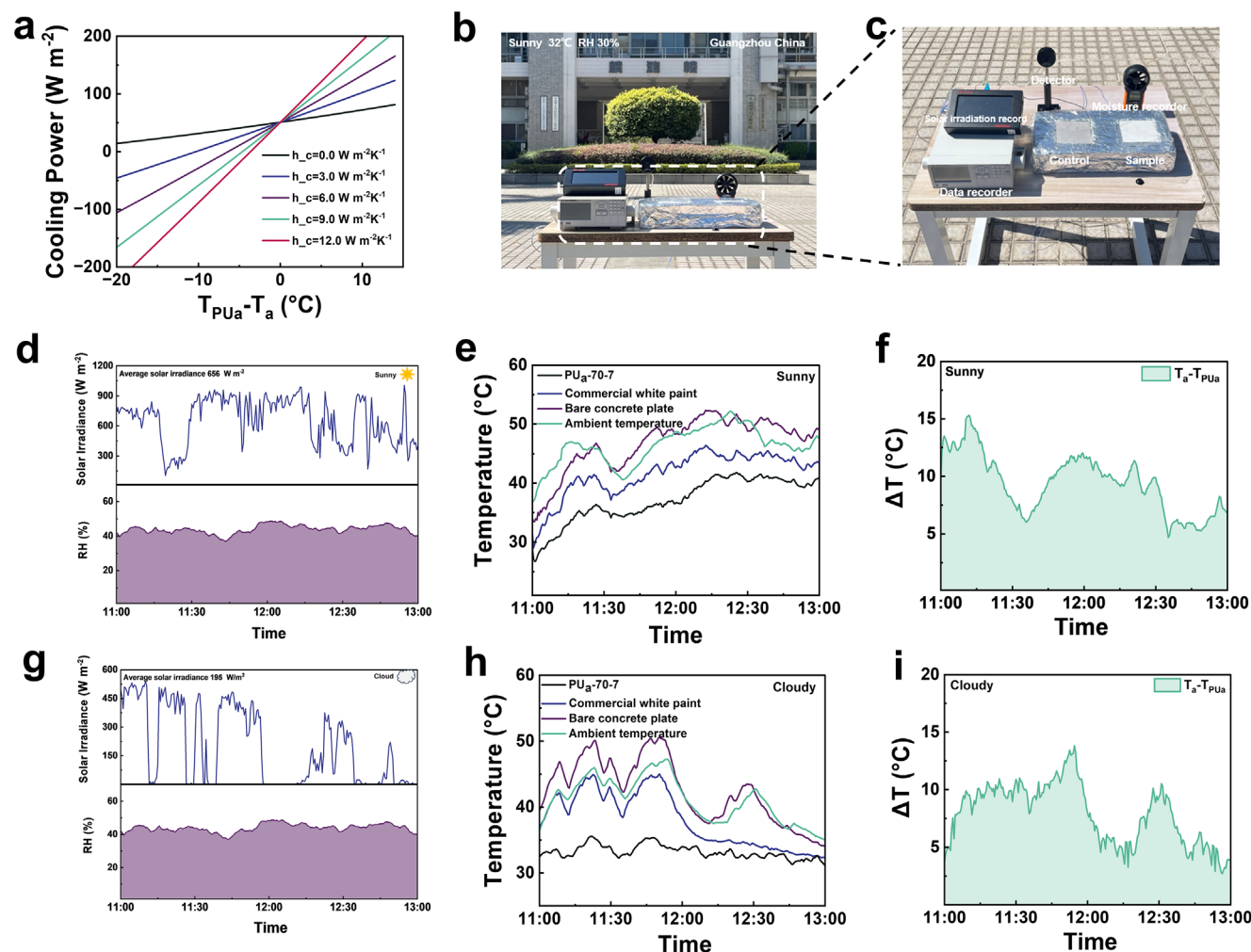


Figure 4. a) Calculated net cooling power of PU_a-70-7 coatings at daytime. b) Photograph and c) schematic of home-built setup for sub-ambient radiative cooling measurements. d–i) Outdoor measurement results between 11:00 and 13:00 in October 2024. (d,g) RH and wind speed. (e,h) Temperature of PU_a-70-7, commercial white paint, bare concrete plate and ambient temperature measured outdoors, respectively. (f,i) The temperature difference between PU_a-70-7 and ambient temperature.

the PDRC coatings exhibit excellent cooling performance under different weather conditions.

2.5. Application

The widespread application of radiative cooling materials is hindered by their limited color choices because most coatings are currently available only in white.^[28] To address this limitation, PU_a-70-7 was mixed with inorganic dyes to enhance its color. Photos of the coatings, which are red, purple, and orange are shown in Figure 5a. Each color exhibits varying degrees of ultraviolet absorption, causing the R_{solar} to reduce in different ways. However, all coatings maintained an average R_{solar} greater than 0.60 (Figure 5b). The orange coating has the highest R_{solar} of 0.77. Indeed, the reflectivity dropped considerably after the addition of pigments (from 0.95 to 0.77), but the emissivity of the coatings showed no significant decrease (Figure 5c). These colored coatings still exhibit a relative high emissivity (≈0.89) compared with

white coatings (≈0.90) in the 8–13 μm atmospheric transmission window. Thus, they still have a certain cooling effect overall. The infrared temperature test results for the colored coatings are shown in Figure S13 (Supporting Information). Although its cooling performance is not as excellent as that of the white coatings, the colored coating still exhibits a noticeable cooling effect (38.1 °C) compared to the blank sample (48.2 °C).

The self-cleaning and antifouling properties are crucial to preventing coating contamination and achieving superior PDRC performance.^[29] The detailed results of the self-cleaning performance tests are presented in Figure 5d. Ink contamination on the coating surface can be easily removed by water-rinsing to restore its original state. After immersing the coating in ink for 10 s, only a minimal residue remained after rinsing, demonstrating its excellent self-cleaning ability. This enhances the environmental resilience of coatings and extends their operational lifespan. To ensure outdoor stability, the adhesion strength of the coating was evaluated using a standard cross-cut tape test (Figure S14, Supporting Information). The results demonstrate minimal square

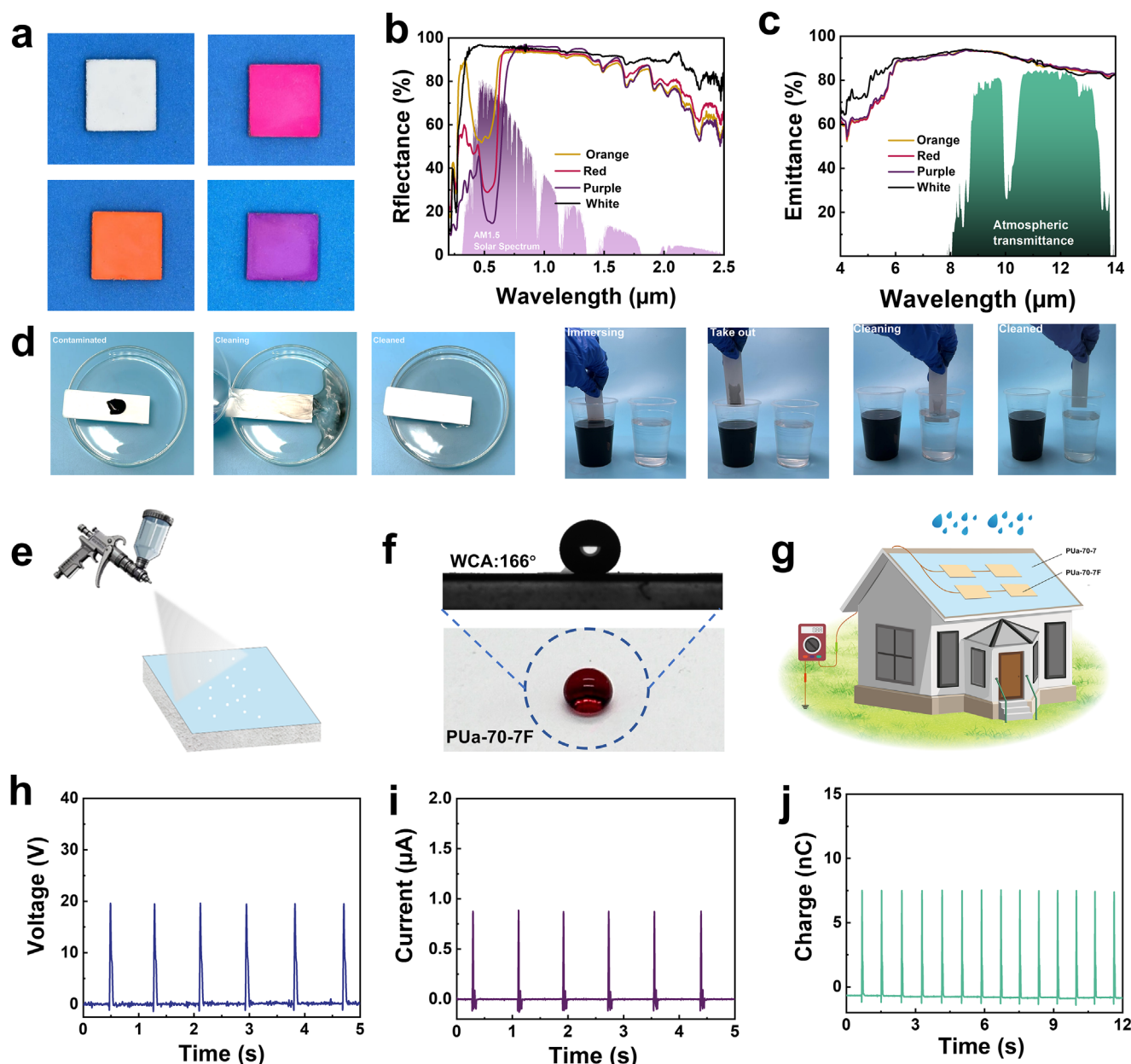


Figure 5. a) Photos of colored coatings. b) Reflectance and c) emittance of colored coatings. d) Self-cleaning capability of the PU_a-70-7 coating contaminated by ink. e) Schematic of sprayed F-SiO₂. f) Water contact angle of PU_a-70-7F. g) Illustration of the performance of water droplet power generation. h–j) Output performances of the PU_a-70-7F (h) V_{OC} , (i) I_{SC} , (j) Q_{TR} .

shedding, indicating that the PU_a-70-7 coating exhibits strong adhesion, with an adhesion rating of 4B.

In addition, the radiative cooling technology can be integrated with water droplet power generation technology. The liquid-solid-based triboelectric nanogenerator (LS-TENG) offers a novel approach to harvesting energy from natural water sources.^[30] The LS-TENG was fabricated by spraying F-SiO₂ on the surface of the PU_a-70-7 coating (Figure 5e). The low surface energy of fluorine and surface micro-nano structure enabled the coating to achieve a WCA of $166^\circ \pm 1^\circ$ (Figure 5f). The coating surface achieved extreme water repellency and demonstrated superhydrophobic properties.^[31] The superhydrophobic surface acceler-

ates droplet sliding and significantly improves droplet separation from the solid surface. The high electronegativity of fluorine facilitates significant charge transfer upon droplet contact, resulting in enhanced voltage and output performance.^[32] When applied to building facades, PU_a-70-7 sprayed with F-SiO₂ (PU_a-70-7F) effectively converts the mechanical energy of falling rainwater into electrical energy (Figure 5g). The output performance of PU_a-70-7F for a single droplet slides down is shown in Figure 5h–j. The cooling effect mitigates excessive evaporation due to solar radiation. These two technologies can be integrated to realize an eco-friendly, efficient, and stable energy-conversion system.

3. Conclusion

In conclusion, we developed a one-component stereoscopic oligosiloxane-based polyurea radiative cooling coating. The coating consists of highly cross-linked organic-inorganic hybrid polyurea and Al_2O_3 particles, demonstrating excellent mechanical properties and outstanding cooling capacity. Compared with current polymer-based PDRC materials, the polyurea network with an inorganic Si—O—Si core and high cross-linking density imparts enhanced modulus and hardness to the coating, ensuring excellent durability in extreme weather conditions. The high infrared emissivity of the Si—O—Si bonds and the high reflectivity of Al_2O_3 enable the coating to achieve an excellent cooling effect. In addition, the coating exhibits outstanding versatility. The introduction of PDMS endows the coating with outstanding self-cleaning properties. Spraying F— SiO_2 on the surface enables superhydrophobicity and water droplet power generation. This facile preparation method can produce PDRC materials on a large scale. This study provides new insights into the design and fabrication of high-performance polymer-based PDRC materials for building cooling spaces.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

mechanical robustness, one-component polyurea, organic-inorganic hybrid, Radiative cooling, Self-cleaning

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